

Ordinary Nonconvex SGD under Distance-Dependent Moments

Finite-Horizon Stationarity and Nagaev Bounds

Wei Biao Wu

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Abstract

Uniform noise-moment bounds exclude stochastic gradients whose variability increases with the iterate. We study ordinary, single-sample stochastic gradient descent for smooth, lower-bounded, possibly nonconvex objectives under distance-dependent conditional moments. Under second moments alone, a direct descent-displacement argument yields $T^{-1/3}$ expected average squared-gradient stationarity with a horizon-dependent stepsize. An explicit oracle-complexity corollary matches the known smooth Blum-Gladyshev (BG-0) lower bound, including the $Lb_2\Delta^3\epsilon^{-6}$ and $L\Delta\sigma^2\epsilon^{-4}$ stochastic terms, where Δ is the initial objective gap and $\sigma^2 + b_2\|x - x_1\|^2$ bounds the variance. Thus unchanged SGD attains the minimax stochastic complexity in this second-moment class. For $p > 2$, predictable localization and a Hilbert-space Fuk-Nagaev inequality yield a high-probability bound separating logarithmic variance and polynomial rare-shock contributions. The localization radius is derived from the recursion: no bounded-iterate assumption, clipping, normalization, momentum, or increasing batch size is needed. We also give increasing-confidence rates, an objective-gap-growth refinement recovering root- T stationarity, and stochastic L^p -Lipschitz examples. The broad BG-0 optimality statement is distinguished from the smaller mean-square-smooth class, in which additional oracle structure permits faster algorithms.

Keywords: stochastic gradient descent; nonconvex optimization; distance-dependent variance; stochastic Lipschitz continuity; martingale concentration; Fuk-Nagaev inequality.

1 Introduction

The stochastic gradient available at a parameter x need not have a noise distribution whose scale is bounded independently of x . Random-design least squares provides a simple example: its gradient contains a random matrix multiplied by the current parameter. More generally, an L^p -Lipschitz stochastic-gradient field has noise moments that can grow linearly with the distance from a reference point. Fresh independent observations guarantee conditional unbiasedness along an adaptive optimization trajectory, but do not turn this state-dependent scale into a deterministic uniform bound.

Stochastic gradient methods are central to large-scale learning (Bottou et al., 2018). This distinction matters for nonconvex optimization. In a strongly convex model, control of objective values or distance to a minimizer may supply the missing stability. For a general smooth, lower-bounded objective, a small objective value need not control the distance traveled, and the objective

may be constant in directions along which the stochastic-gradient variance increases. A proof that simply substitutes the moment at initialization for a global noise envelope is therefore invalid.

We consider the unmodified recursion

$$x_{t+1} = x_t - \alpha G(x_t, Z_t), \quad t = 1, \dots, T, \quad (1.1)$$

with one fresh independent observation Z_t at each iteration. The stepsize is constant within a run and may be selected as a function of its horizon and desired confidence. Our principal criterion is

$$A_T = \frac{1}{T} \sum_{t=1}^T \|\nabla F(x_t)\|^2, \quad F(x) = \mathbb{E}f(x, Z) \quad (1.2)$$

when a differentiable random loss representation is available. The results require an unbiased stochastic-gradient oracle, not an interchange of differentiation and expectation without justification. We also state the theorems directly for martingale-difference errors, so that the exact role of independent sampling is transparent.

The main assumption bounds the conditional second moment and, for the concentration result, the squared conditional L^p norm of the error by an affine function of $\|x_t - x_1\|^2$. A two-point stochastic Lipschitz assumption is sufficient, but is not needed by the optimization argument. This distinction enlarges the theorem's scope and identifies precisely what enters the proof.

1.1 Results and mechanism

Our first result uses only conditional second moments. It couples expected descent with the exact displacement decomposition

$$x_{k+1} - x_1 = -\alpha \sum_{t=1}^k \nabla F(x_t) - \alpha \sum_{t=1}^k \xi_t, \quad \xi_t = G(x_t, Z_t) - \nabla F(x_t). \quad (1.3)$$

The deterministic gradient sum is controlled by Cauchy–Schwarz, while martingale orthogonality controls the second moment of the noise sum. The resulting feedback term can be absorbed when a quantity proportional to

$$h_T = \alpha^2 T(1 + L\alpha T) \quad (1.4)$$

is sufficiently small. We obtain the familiar bounded-variance form, up to a factor of two from the absorption step:

$$\mathbb{E}A_T \leq \frac{4\Delta}{\alpha T} + 2L\alpha\sigma^2, \quad \Delta = F(x_1) - F_*,$$

but with an additional horizon-dependent admissibility condition. Here σ^2 is an intercept in the distance-dependent variance bound, not a uniform variance bound over $\mathbb{R}^d$. A step $\alpha \asymp T^{-2/3}$ makes the feedback condition hold and yields $\mathbb{E}A_T = O(T^{-1/3})$. More precisely, Corollary 3.4 gives a deterministic oracle budget of order

$$1 + \frac{L\Delta}{\varepsilon^2} + \frac{L\Delta\sigma^2}{\varepsilon^4} + \frac{Lb_2\Delta^3}{\varepsilon^6} \quad (1.5)$$

in the small-accuracy regime, for a uniformly randomized output satisfying $\mathbb{E}\|\nabla F(\hat{x})\|^2 \leq \varepsilon^2$. The two stochastic terms match the lower bound of Fazla et al. (2026, Theorem 1); Section 3.1 gives the precise oracle-class and accuracy comparison.

Our second result refines the confidence dependence. A Hilbert-space martingale inequality controls both the accumulated squared errors and the maximum noise partial sum. An objective-level stop controls the adaptive martingale transform in the descent inequality. A second, distance-based stop produces a deterministically bounded noise envelope for an auxiliary recursion. On a single high-probability event, an explicit radius prevents either recursion from leaving the localization region. Consequently the auxiliary recursion agrees with the original SGD recursion throughout the run. No projection or stopping rule is implemented by the optimizer.

The resulting bound separates

$$L\alpha\sigma^2\log(6/\delta) \quad \text{and} \quad L\alpha\nu_p^2T^{2/p-1}\delta^{-2/p}.$$

The first term reflects a variance scale and the second a finite- p rare-shock scale. Both the confidence level and the growth coefficients enter the stepsize restriction. At fixed problem parameters and confidence, the rate is $T^{-1/3}$. For example, $p = 4$ permits a bound of order $T^{-1/3}(\log T)^{1/3}$ with failure probability T^{-1} after a logarithmic stepsize adjustment. These are finite-horizon assertions; different horizons may use different stepsizes and therefore different runs.

A separate objective-gap-growth assumption allows the same localization idea to close with $L\alpha^2T$ rather than h_T , recovering a root- T guarantee. Examples distinguish this additional geometry from stochastic Lipschitz continuity alone. Finally, an additive-shock calculation records why a polynomial confidence contribution cannot generally be removed for unchanged SGD under finite moments.

1.2 Relation to existing work

Distance growth and fixed-horizon analysis. Distance-dependent moment conditions originate in stochastic approximation (Blum, 1954; Gladyshev, 1965). The BG-0 terminology records that the reference point is the initialization. Alacaoglu et al. (2026) develop convex guarantees using Halpern anchoring and compare these growth assumptions with expected smoothness. Centered-noise and uncentered-gradient conditions must be distinguished; Proposition 2.8 gives their relation under objective smoothness. For convex composite optimization, Domke et al. (2023, Theorem 8) already use a fixed horizon to control quadratically growing estimator moments. The difficulty of a direct nonconvex analysis, without a regularization device, is identified in Section 5 of Alacaoglu et al. (2026). Theorem 3.1 answers its fixed-horizon version; a horizon-free, single-run result is a separate question.

Oracle complexity and the role of the update. Ghadimi and Lan (2013) give foundational finite-variance nonconvex stationarity guarantees. The bounded-variance oracle lower bounds of Arjevani et al. (2023) distinguish objective smoothness from mean-square smoothness of the stochastic oracle. Under BG-0, Fazla et al. (2026) establish the corresponding ε^{-6} lower bound and achieve this order with distance-dependent dynamic batching. Upadhyay et al. (2026, Corollary 3.1) obtain the same order for an expected gradient-norm criterion using normalized momentum. Our second-moment result attains the BG-0 stochastic complexity with exactly one fresh oracle call and the ordinary SGD update at every iteration. The direct descent-displacement closure, rather than a new growth assumption or a new worst-case exponent, is the algorithm-specific contribution.

Additional geometry and stochastic smoothness. The objective-gap second-moment condition in Section 8 lies in the expected-smoothness/ABC framework of Khaled and Richtárik (2023); our result there is a finite- p confidence refinement. Adaptive methods under gradient-dependent affine variance, such as the AdaGrad-Norm analysis of Faw et al. (2022), use a different growth condition and update. The stochastic L^p -Lipschitz condition of Li et al. (2026, Assumption 2 in the public draft) supplies our distance-growth assumptions from a moment at one point. Their moment-contraction analysis also uses strong convexity and a noise-sensitive stepsize. Constant-step invariant-law and averaged-iterate analyses under additional geometry appear in Dieuleveut et al. (2020). For $p > 2$, stochastic L^p -Lipschitz continuity implies mean-square smoothness. The faster variance-reduced BG-0 methods of Fazla et al. (2026) exploit that extra structure, so the broad-class optimality result is not an optimality assertion for this smaller oracle class.

Heavy tails and high-probability optimization. Zhang et al. (2020) connect heavy-tailed stochastic gradients with attention models and establish optimization upper and lower bounds in a finite- q regime with $1 < q \leq 2$. Their Theorem 6 gives an $\Omega(\varepsilon^{-(3q-2)/(q-1)})$ nonconvex query lower bound under their uniform central- q -moment assumptions and stated parameter regime; this differs from the distance-growth, finite-variance BG-0 lower bound. Clipping-based high-probability analyses include convex accelerated methods (Gorbunov et al., 2020), clipping with normalization and momentum for nonconvex objectives (Cutkosky and Mehta, 2021), and finite-central-moment results (Sadiev et al., 2023; Nguyen et al., 2023). Liu and Zhou (2025) obtain clipping-free heavy-tail guarantees using batched normalized momentum, not the ordinary update. Under sub-Gaussian noise, Liu et al. (2023) give high-probability results for SGD and adaptive methods. The cited robust methods can obtain logarithmic confidence dependence under their respective uniform moment assumptions, whereas our unchanged update retains a polynomial rare-shock term. These works separate two issues that also matter here: an infinite variance caused by $q < 2$ is different from finite conditional variance that is unbounded over the parameter space, and robust updates have different confidence costs from unchanged SGD. Our $p > 2$ theorem keeps the ordinary update and separates a variance-scale contribution from a polynomial rare-shock contribution under distance-dependent moments.

Probability tools and Nagaev bounds for SGD. The exponential-plus-polynomial approach originates with Fuk and Nagaev (1971); Nagaev (1979). Martingale developments include Pinelis (1994); Fan et al. (2017); Rio (2017); the sole non-elementary concentration input used below is the maximal Hilbert-space specialization of Mollenhauer and Fiedler (2025, Theorem 2.2). For averaged SGD in a linear model, Zhu et al. (2022) already obtain Nagaev estimation bounds and demonstrate the necessity of a polynomial confidence contribution. Our optimization criterion is instead nonconvex average squared-gradient stationarity. The new extension derives deterministic conditional envelopes from iterate-dependent growth by closing the displacement bound, rather than assuming bounded iterates. Section 11.1 records the supplied unpublished source of the deterministic-envelope and quadratic-shock ingredients, which are included with full proofs and are not counted among the new results.

Organization. Section 2 states the assumptions and verifies stochastic Lipschitz oracles. Section 3 proves the second-moment theorem. Sections 4 and 5 develop the probability and deterministic-envelope ingredients. Section 6 proves the distance-dependent Nagaev theorem, and Section 7 gives

its stepsize and confidence consequences. Section 8 treats objective-gap growth. Sections 9 and 10 give examples, a stability counterexample, and the additive rare-shock calculation. Section 11 records the scope and limitations.

2 Setup and moment assumptions

Let $(\Omega, \mathcal{F}, \mathbb{P})$ carry an increasing filtration $(\mathcal{F}_t)_{t=0}^T$. Write $\mathbb{E}_{t-1}Y = \mathbb{E}(Y \mid \mathcal{F}_{t-1})$. Norms without a subscript are Euclidean, matrix operator norms are denoted by $\|\cdot\|_{\text{op}}$, and $\|Y\|_q = (\mathbb{E}\|Y\|^q)^{1/q}$. All almost-sure conditional inequalities are understood for the finitely many indices in the run. Conditional moments of nonnegative variables are initially interpreted in the extended sense; Lemma 2.6 then establishes the finite-horizon integrability used in the proofs.

Assumption 2.1 (Objective and recursion). The function $F : \mathbb{R}^d \rightarrow \mathbb{R}$ is differentiable, its gradient is L -Lipschitz for some $L > 0$, and $F(x) \geq F_*$ for a finite constant F_* . The initialization x_1 is deterministic. For a deterministic stepsize $\alpha > 0$,

$$x_{t+1} = x_t - \alpha(g_t + \xi_t), \quad g_t = \nabla F(x_t), \quad 1 \leq t \leq T, \quad (2.1)$$

where x_t is $\mathcal{F}_{t-1}$ -measurable, ξ_t is $\mathcal{F}_t$ -measurable, and $\mathbb{E}_{t-1}\xi_t = 0$.

We use $\Delta = F(x_1) - F_*$ and $A_T = T^{-1} \sum_{t=1}^T \|g_t\|^2$ throughout. The lower bound F_* need not be the attained minimum, and no set of minimizers is required. Every upper bound remains valid when Δ is replaced by a known deterministic upper bound on $F(x_1) - F_*$.

Assumption 2.2 (Distance-dependent second moment). There are finite $\sigma^2, b_2 \geq 0$ such that

$$\mathbb{E}_{t-1} \|\xi_t\|^2 \leq \sigma^2 + b_2 \|x_t - x_1\|^2 \quad \text{almost surely,} \quad 1 \leq t \leq T. \quad (2.2)$$

Assumption 2.3 (Distance-dependent finite- p moments). For a fixed $p > 2$, Assumption 2.2 holds and there are finite $\nu_p^2, b_p \geq 0$ such that

$$(\mathbb{E}_{t-1} \|\xi_t\|^p)^{2/p} \leq \nu_p^2 + b_p \|x_t - x_1\|^2 \quad \text{almost surely,} \quad 1 \leq t \leq T. \quad (2.3)$$

The quantities b_2, b_p multiply a squared distance. The quantity ν_p^2 is a squared L^p scale; it is not a p th moment. Both intercepts may be chosen as upper bounds rather than exact moments. The constants are deterministic, and rate statements hold them fixed as the horizon increases.

2.1 Independent samples and stochastic Lipschitz continuity

Let Z_t be iid with law P_Z , independent of $\mathcal{F}_{t-1}$, and let $G : \mathbb{R}^d \times \mathcal{Z} \rightarrow \mathbb{R}^d$ be jointly measurable. Suppose, for every deterministic x ,

$$\mathbb{E}_Z G(x, Z) = \nabla F(x), \quad e(x, z) = G(x, z) - \nabla F(x). \quad (2.4)$$

The oracle representation in (2.1) is $\xi_t = e(x_t, Z_t)$.

Proposition 2.4 (Verification from stochastic Lipschitz continuity). *Suppose that for $p > 2$,*

$$\|G(x_1, Z)\|_p < \infty, \quad \|G(x, Z) - G(y, Z)\|_p \leq L_{G,p} \|x - y\| \quad (x, y \in \mathbb{R}^d). \quad (2.5)$$

For $q = 2, p$, let $L_{G,q}$ be an admissible L^q -Lipschitz constant, put

$$s_q = \|e(x_1, Z)\|_q, \quad K_q = L_{G,q} + L,$$

and take $L_{G,2} = L_{G,p}$ when no sharper second-moment constant is available. Then

$$\|e(x, Z) - e(y, Z)\|_q \leq K_q \|x - y\|, \quad \|e(x, Z)\|_q \leq s_q + K_q \|x - x_1\|. \quad (2.6)$$

Assumption 2.3 holds with

$$\sigma^2 = 2s_2^2, \quad b_2 = 2K_2^2, \quad \nu_p^2 = 2s_p^2, \quad b_p = 2K_p^2. \quad (2.7)$$

The finite moment in (2.5) may instead be imposed at any deterministic reference point.

Proof. Monotonicity of probability-space L^q norms gives the second-moment Lipschitz inequality from the p th-moment inequality. The triangle inequality and objective smoothness give

$$\|e(x, Z) - e(y, Z)\|_q \leq \|G(x, Z) - G(y, Z)\|_q + \|\nabla F(x) - \nabla F(y)\| \leq K_q \|x - y\|.$$

Set $y = x_1$ and use the triangle inequality once more to obtain (2.6). Squaring and using $(a + b)^2 \leq 2a^2 + 2b^2$ gives the pointwise moment-growth bounds.

Independence and joint measurability imply, for predictable x_t ,

$$\mathbb{E}_{t-1} e(x_t, Z_t) = 0, \quad \mathbb{E}_{t-1} \|e(x_t, Z_t)\|^q = \mathbb{E}_Z \|e(x, Z)\|^q|_{x=x_t}. \quad (2.8)$$

The identity for nonnegative functions follows first for product indicator functions and then by the monotone class theorem. For vector means it follows by integrability and componentwise application. Finite-horizon integrability also follows by the induction in Lemma 2.6. Substitution into the pointwise inequalities proves (2.7). Finally,

$$\|G(x_1, Z)\|_p \leq \|G(x_{\text{ref}}, Z)\|_p + L_{G,p} \|x_1 - x_{\text{ref}}\|$$

transfers any finite reference-point moment to the initialization. $\square$

Remark 2.5. The quantifiers in (2.5) do not imply $\sup_x \|e(x, Z)\|_p < \infty$. They control increments of the random field, not its absolute size over an unbounded parameter space. Nor does fresh-sample independence remove $\|x_t - x_1\|$ from (2.8). In addition, a two-point increment condition alone does not imply a finite moment at any point: an arbitrary additive random vector can be added without changing the increments. The single-point moment condition is indispensable for this verification.

2.2 Integrability and elementary objective bounds

Lemma 2.6 (Finite-horizon integrability). *Under Assumptions 2.1 and 2.2, x_t, g_t, ξ_t are square-integrable for every finite index in the run, and $F(x_t)$ is integrable. Under Assumption 2.3, these vectors belong to L^p as well.*

Proof. Write $d_t = x_t - x_1$. Objective smoothness gives

$$\|g_t\| \leq \|\nabla F(x_1)\| + L \|d_t\|. \quad (2.9)$$

For $q = 2$ use $\omega_q = \sigma^2, c_q = b_2$, and for $q = p$ use $\omega_q = \nu_p^2, c_q = b_p$. The conditional growth inequality and Minkowski give

$$\begin{aligned}\|\xi_t\|_q &= \left\| (\mathbb{E}_{t-1} \|\xi_t\|^q)^{1/q} \right\|_q \\ &\leq \left\| \sqrt{\omega_q + c_q} \|d_t\|^2 \right\|_q \leq \sqrt{\omega_q} + \sqrt{c_q} \|d_t\|_q.\end{aligned}$$

Starting with $d_1 = 0$, the recursion implies

$$\|d_{t+1}\|_q \leq \{1 + \alpha(L + \sqrt{c_q})\} \|d_t\|_q + \alpha\{\|\nabla F(x_1)\| + \sqrt{\omega_q}\}.$$

An induction proves finiteness at every finite time. No uniform-in- t bound is asserted by this induction. The lower bound on F and the smooth upper estimate

$$F(x_t) \leq F(x_1) + \langle \nabla F(x_1), d_t \rangle + \frac{L}{2} \|d_t\|^2$$

then show $F(x_t) \in L^1$. □

Lemma 2.7 (Self-bounding gradient). *For every $x \in \mathbb{R}^d$,*

$$\|\nabla F(x)\|^2 \leq 2L\{F(x) - F_*\}. \quad (2.10)$$

Proof. The descent lemma at $y = x - L^{-1}\nabla F(x)$ gives

$$F_* \leq F(y) \leq F(x) - L^{-1} \|\nabla F(x)\|^2 + (2L)^{-1} \|\nabla F(x)\|^2.$$

Rearrangement proves the claim. Attainment of F_* is not used. □

Proposition 2.8 (Centered and uncentered distance growth). *For an unbiased iid oracle, suppose*

$$\mathbb{E}_Z \|e(x, Z)\|^2 \leq \sigma^2 + b \|x - x_1\|^2.$$

Then

$$\mathbb{E}_Z \|G(x, Z)\|^2 \leq \sigma^2 + 2 \|\nabla F(x_1)\|^2 + (b + 2L^2) \|x - x_1\|^2. \quad (2.11)$$

Conversely, an uncentered bound $\mathbb{E}_Z \|G(x, Z)\|^2 \leq a + c \|x - x_1\|^2$ implies the same bound for $\mathbb{E}_Z \|e(x, Z)\|^2$.

Proof. Unbiasedness gives the orthogonal decomposition

$$\mathbb{E}_Z \|G(x, Z)\|^2 = \|\nabla F(x)\|^2 + \mathbb{E}_Z \|e(x, Z)\|^2.$$

The gradient estimate (2.9), with $(a + b)^2 \leq 2a^2 + 2b^2$, proves (2.11). Dropping the nonnegative squared-gradient term proves the converse. □

3 A second-moment stationarity theorem

The first theorem requires no p th moment with $p > 2$. Its purpose is to separate the finite-horizon stability issue from the additional concentration issue.

Theorem 3.1 (Expected stationarity and marginal displacement). *Under Assumptions 2.1 and 2.2, define*

$$h_T = \alpha^2 T(1 + L\alpha T). \quad (3.1)$$

If

$$0 < \alpha \leq L^{-1}, \quad b_2 h_T \leq \frac{1}{4}, \quad (3.2)$$

then

$$\boxed{\mathbb{E} A_T \leq \frac{4\Delta}{\alpha T} + 2L\alpha\sigma^2.} \quad (3.3)$$

Moreover,

$$\max_{1 \leq t \leq T+1} \mathbb{E} \|x_t - x_1\|^2 \leq 8\alpha T\Delta + 4h_T\sigma^2. \quad (3.4)$$

The maximum in (3.4) is outside the expectation.

Proof. All expectations below are justified by Lemma 2.6. Define

$$S_T = \sum_{t=1}^T \mathbb{E} \|g_t\|^2, \quad V_T = \sum_{t=1}^T \mathbb{E} \|\xi_t\|^2, \quad R_T = \max_{1 \leq t \leq T+1} \mathbb{E} \|x_t - x_1\|^2.$$

The smoothness inequality and conditional centering imply

$$\begin{aligned} \mathbb{E}_{t-1} F(x_{t+1}) &\leq F(x_t) - \alpha \|g_t\|^2 + \frac{L\alpha^2}{2} \mathbb{E}_{t-1} \|g_t + \xi_t\|^2 \\ &= F(x_t) - \alpha \left(1 - \frac{L\alpha}{2}\right) \|g_t\|^2 + \frac{L\alpha^2}{2} \mathbb{E}_{t-1} \|\xi_t\|^2. \end{aligned} \quad (3.5)$$

Since $\alpha \leq L^{-1}$, summation, total expectation, and $F(x_{T+1}) \geq F_*$ give

$$S_T \leq \frac{2\Delta}{\alpha} + L\alpha V_T. \quad (3.6)$$

For $k \leq T$, the exact identity (1.3) and $(a+b)^2 \leq 2a^2 + 2b^2$ yield

$$\mathbb{E} \|x_{k+1} - x_1\|^2 \leq 2\alpha^2 k \sum_{t=1}^k \mathbb{E} \|g_t\|^2 + 2\alpha^2 \mathbb{E} \left\| \sum_{t=1}^k \xi_t \right\|^2.$$

For $s < t$, ξ_s is $\mathcal{F}_{t-1}$ -measurable and $\mathbb{E} \langle \xi_s, \xi_t \rangle = \mathbb{E} \langle \xi_s, \mathbb{E}_{t-1} \xi_t \rangle = 0$. Thus martingale orthogonality gives

$$R_T \leq 2\alpha^2 T S_T + 2\alpha^2 V_T. \quad (3.7)$$

Taking expectations in (2.2) also gives

$$V_T \leq T(\sigma^2 + b_2 R_T). \quad (3.8)$$

Combining (3.6)–(3.8),

$$R_T \leq 4\alpha T\Delta + 2h_T(\sigma^2 + b_2 R_T). \quad (3.9)$$

As $2b_2h_T \leq 1/2$, the last term can be moved to the left, proving (3.4).

Substituting this displacement bound into (3.6) and (3.8) gives

$$\mathbb{E}A_T \leq \frac{2\Delta}{\alpha T} + L\alpha\sigma^2 + 8Lb_2\alpha^2T\Delta + 4Lb_2\alpha h_T\sigma^2. \quad (3.10)$$

The inequality $L\alpha^3T^2 \leq h_T$ implies

$$8Lb_2\alpha^2T\Delta \leq 8b_2h_T\frac{\Delta}{\alpha T} \leq \frac{2\Delta}{\alpha T}, \quad 4Lb_2\alpha h_T\sigma^2 \leq L\alpha\sigma^2.$$

These two estimates prove (3.3). $\square$

Remark 3.2 (An unabsorbed version). When $2b_2h_T < 1$, the proof directly yields

$$R_T \leq \frac{4\alpha T\Delta + 2h_T\sigma^2}{1 - 2b_2h_T}, \quad \mathbb{E}A_T \leq \frac{2\Delta}{\alpha T} + L\alpha(\sigma^2 + b_2R_T).$$

The constants in Theorem 3.1 result from the convenient sufficient restriction $b_2h_T \leq 1/4$. They are not claimed to be best possible.

Corollary 3.3 (A one-sample $T^{-1/3}$ expectation rate). *Fix the objective and the constants in Assumption 2.2. Choose $a_0 > 0$ such that*

$$a_0 \leq L^{-1}, \quad b_2(a_0^2 + La_0^3) \leq \frac{1}{4}, \quad (3.11)$$

and set $\alpha = a_0T^{-2/3}$. Then for every positive integer T ,

$$\mathbb{E}A_T \leq \frac{4\Delta}{a_0}T^{-1/3} + 2La_0\sigma^2T^{-2/3}. \quad (3.12)$$

At fixed $\delta \in (0, 1)$ the same right side divided by δ is a valid $1 - \delta$ probability bound.

Proof. Here $h_T = a_0^2T^{-1/3} + La_0^3 \leq a_0^2 + La_0^3$ and $\alpha \leq a_0 \leq L^{-1}$. Apply Theorem 3.1 and then Markov's inequality to the nonnegative variable A_T . $\square$

A fully explicit admissible choice is any positive a_0 no larger than

$$\min \left\{ L^{-1}, (8b_2)^{-1/2}, (8Lb_2)^{-1/3} \right\},$$

with the two growth restrictions omitted if $b_2 = 0$. The absence of b_2 from the displayed noise term in (3.3) does not mean that noise growth is free: it enters the admissible stepsize through (3.2).

3.1 Oracle complexity and BG-0 optimality

A power law with all parameters hidden does not identify the variance-growth cost. The next corollary makes the comparison quantitative and uses no higher moments.

Corollary 3.4 (Parameter-explicit one-sample oracle complexity). *Assume Assumptions 2.1 and 2.2, let $\Delta > 0$, and fix $\varepsilon > 0$. Choose an integer T satisfying*

$$T \geq \max \left\{ 1, \frac{8L\Delta}{\varepsilon^2}, \frac{32L\Delta\sigma^2}{\varepsilon^4}, \frac{512b_2\Delta^2}{\varepsilon^4}, \frac{4096Lb_2\Delta^3}{\varepsilon^6} \right\}, \quad (3.13)$$

and set $\alpha = 8\Delta/(T\varepsilon^2)$. Then $\mathbb{E}A_T \leq \varepsilon^2$. If τ is uniform on $\{1, \dots, T\}$ and independent of the run, $\hat{x} = x_\tau$ satisfies

$$\mathbb{E}\|\nabla F(\hat{x})\|^2 \leq \varepsilon^2, \quad \mathbb{E}\|\nabla F(\hat{x})\| \leq \varepsilon, \quad (3.14)$$

using exactly T stochastic-gradient queries. If $\varepsilon^2 \leq L\Delta$, the sufficient budget has order

$$T = O \left(1 + \frac{L\Delta}{\varepsilon^2} + \frac{L\Delta\sigma^2}{\varepsilon^4} + \frac{Lb_2\Delta^3}{\varepsilon^6} \right), \quad (3.15)$$

with an absolute implied constant. Full-gradient evaluations are not needed to select the randomized output.

Proof. The smoothness restriction follows from $T \geq 8L\Delta/\varepsilon^2$. Direct substitution gives

$$b_2h_T = \frac{64b_2\Delta^2}{T\varepsilon^4} + \frac{512Lb_2\Delta^3}{T\varepsilon^6} \leq \frac{1}{8} + \frac{1}{8} = \frac{1}{4}.$$

Theorem 3.1 therefore applies. Its two stationarity terms satisfy

$$\frac{4\Delta}{\alpha T} = \frac{\varepsilon^2}{2}, \quad 2L\alpha\sigma^2 = \frac{16L\Delta\sigma^2}{T\varepsilon^2} \leq \frac{\varepsilon^2}{2}.$$

Conditionally on the iterates, $\mathbb{E}(\|\nabla F(x_\tau)\|^2 \mid x_1, \dots, x_T) = A_T$. Taking expectations and applying Cauchy–Schwarz proves (3.14). Finally, if $\varepsilon^2 \leq L\Delta$, then

$$\frac{b_2\Delta^2}{\varepsilon^4} \leq \frac{Lb_2\Delta^3}{\varepsilon^6}.$$

Choose T as the ceiling of the maximum in (3.13), and bound that maximum by a sum to obtain (3.15). $\square$

The lower-bound comparison. Consider the pointwise iid-oracle subclass in which F is L -smooth, $F(x_1) - \inf F \leq \Delta$, and

$$\mathbb{E}_Z \|G(x, Z) - \nabla F(x)\|^2 \leq \sigma^2 + b_2\|x - x_1\|^2 \quad (x \in \mathbb{R}^d). \quad (3.16)$$

The dimension is unrestricted. With the identification $B_v^2 = b_2$, $b_v^2 = \sigma^2$, and $x_0 = x_1$, Fazla et al. (2026, Theorem 1) give the lower bound

$$\Omega \left(\frac{Lb_2\Delta^3}{\varepsilon^6} + \frac{L\Delta\sigma^2}{\varepsilon^4} \right) \quad (3.17)$$

on the expected number of queries of any randomized algorithm achieving $\mathbb{E}\|\nabla F(\hat{x})\| \leq \varepsilon$. Their theorem assumes $L, \Delta > 0$ and $\varepsilon^2 \leq L\Delta/(1536\ell_1)$ for a numerical constant $\ell_1 > 0$. A lower bound for this first-moment criterion also applies to the stronger squared-gradient criterion in (3.14). Our deterministic query budget is, in particular, an expected query budget.

Thus (3.15) matches both stochastic terms of (3.17), including their dependence on L, b_2, Δ , and σ^2 , with the usual deterministic $L\Delta/\varepsilon^2$ overhead. In the small-accuracy stochastic regime where these terms dominate, ordinary single-sample SGD is minimax optimal in the smooth BG-0 class. In particular, for fixed $b_2 > 0$ and $L, \Delta > 0$, its worst-case order is $\Theta(\varepsilon^{-6})$. When $b_2 = 0$, the same corollary reduces to the bounded-variance ε^{-4} stochastic complexity. This comparison uses the cited lower-bound theorem; it is not a new oracle lower-bound proof.

Why the oracle class matters. The growth condition (3.16) compares each query with the initialization. It imposes no common-sample two-point regularity. By contrast,

$$\|G(x, Z) - G(y, Z)\|_2 \leq \|G(x, Z) - G(y, Z)\|_p \leq L_{G,p}\|x - y\|$$

shows that stochastic L^p -Lipschitz continuity with $p > 2$ implies mean-square smoothness. Theorems 2 and 7 of Fazla et al. (2026) distinguish that more structured oracle model and give an ε^{-4} worst-case exponent under their stated parameter restrictions, with variance-reduced upper bounds. The ε^{-6} BG-0 optimality claim therefore cannot be restricted automatically to that smaller class. Its best achievable rate for the unchanged single-sample SGD recursion remains a separate algorithm-specific question.

4 The probability input and predictable stopping

The only non-elementary concentration input is the following maximal inequality. The maximal form and its constants are already supplied by Mollenhauer and Fiedler (2025, Theorem 2.2); no additional maximal-inequality result is claimed here. Hilbert spaces have smoothness constant $D = 1$, independently of dimension.¹

Theorem 4.1 (Maximal Hilbert-space Fuk–Nagaev inequality). (Mollenhauer and Fiedler, 2025, Theorem 2.2, Hilbert-space specialization). *Let $p > 2$ and let $X_1, \dots, X_T$ be martingale differences in a separable Hilbert space. If deterministic $v, m_p \geq 0$ satisfy*

$$\sum_{t=1}^T \mathbb{E}_{t-1} \|X_t\|^2 \leq v, \quad \sum_{t=1}^T \mathbb{E}_{t-1} \|X_t\|^p \leq m_p \quad \text{almost surely,}$$

then, for every $u \in (0, 1)$, with probability at least $1 - u$,

$$\max_{1 \leq k \leq T} \left\| \sum_{t=1}^k X_t \right\| \leq \sqrt{2v \log(2/u)} + \mathfrak{c}_p (2m_p/u)^{1/p}, \quad (4.1)$$

Here

$$\mathfrak{c}_p = \frac{1}{2p} + \min\{1/p, 1/5\} + 1 + \mathbf{1}_{\{p>3\}} \frac{p}{3}.$$

¹Equation (2.2) of the cited version has the index slip $\max_i \|M_n\|$. We use the intended $\max_i \|M_i\|$, consistent with its maximal formulation and the independent-sum specialization (2.5).

The terminal bound follows by taking $k = T$. The budget m_p is a sum of conditional p th moments, not a squared moment scale or a stepsize coefficient.

Lemma 4.2 (Predictable stopping preserves the budgets). *Under the assumptions of Theorem 4.1, let $I_t \in \{0, 1\}$ be $\mathcal{F}_{t-1}$ -measurable. The stopped increments $X_t^\circ = I_t X_t$ satisfy the same maximal bound (4.1) with budgets v, m_p .*

Proof. Conditional centering gives $\mathbb{E}_{t-1} X_t^\circ = 0$, and for $q = 2, p$,

$$\mathbb{E}_{t-1} \|X_t^\circ\|^q = I_t \mathbb{E}_{t-1} \|X_t\|^q \leq \mathbb{E}_{t-1} \|X_t\|^q.$$

Sum and apply Theorem 4.1. For a stopping time τ , the convention $I_t = \mathbf{1}_{\{\tau \geq t\}}$ is predictable and retains the exit-producing increment. No conditioning on a future no-exit event is involved. $\square$

Lemma 4.3 (Raw energy as a Hilbert-space martingale). *Suppose $\xi_t \in \mathbb{R}^d$ are martingale differences with deterministic conditional envelopes*

$$\mathbb{E}_{t-1} \|\xi_t\|^2 \leq s^2, \quad \mathbb{E}_{t-1} \|\xi_t\|^p \leq n_p^p. \quad (4.2)$$

Then with probability at least $1 - u$,

$$\left(\sum_{t=1}^T \|\xi_t\|^2 \right)^{1/2} \leq \sqrt{2Ts^2 \log(2/u)} + \mathfrak{c}_p n_p (2T/u)^{1/p}. \quad (4.3)$$

The same expression bounds the maximum norm of the noise partial sums with probability at least $1 - u$.

Proof. In the Hilbert space $\mathcal{H}_T = (\mathbb{R}^d)^T$, let E_t insert a vector into the t th coordinate and set $X_t = E_t \xi_t$. The map E_t is deterministic and isometric, so (X_t) are Hilbert-valued martingale differences with second- and p th-moment budgets Ts^2 and Tn_p^p . Moreover,

$$\left\| \sum_{t=1}^T E_t \xi_t \right\|_{\mathcal{H}_T}^2 = \sum_{t=1}^T \|\xi_t\|^2.$$

Theorem 4.1 proves (4.3). Apply Theorem 4.1 in $\mathbb{R}^d$ to obtain the partial-sum assertion. These are two separate probability statements; simultaneous use requires a union bound. $\square$

5 A deterministic-envelope lemma for plain SGD

This section proves the deterministic-envelope optimization ingredient directly from Theorem 4.1. The raw-energy lift and objective-level localization are auxiliary ingredients, not new concentration theorems. Working directly with plain SGD permits the smoothness cap $\alpha \leq L^{-1}$. The relation to the supplied unpublished manuscript is recorded in Section 11.1.

For $p > 2$, define the numerical constant

$$\mathfrak{D}_p = 4 + 2\mathfrak{c}_p^2 6^{2/p}. \quad (5.1)$$

For a horizon and failure probability, use

$$\ell = \log(6/\delta), \quad \rho = T^{2/p-1} \delta^{-2/p}. \quad (5.2)$$

Proposition 5.1 (Uniform envelopes: simultaneous trajectory bounds). *Under Assumption 2.1, suppose (4.2) holds and $0 < \alpha \leq L^{-1}$. Put*

$$J = \sqrt{2Ts^2\ell} + \mathfrak{c}_p n_p (6T/\delta)^{1/p}, \quad H = 4\Delta + 6L\alpha^2 J^2. \quad (5.3)$$

With probability at least $1 - \delta$, the following four inequalities hold simultaneously:

$$\sum_{t=1}^T \|\xi_t\|^2 \leq J^2, \quad \max_{k \leq T} \left\| \sum_{t=1}^k \xi_t \right\| \leq J, \quad (5.4)$$

$$\max_{1 \leq t \leq T+1} \{F(x_t) - F_*\} \leq H, \quad A_T \leq \frac{4\Delta}{\alpha T} + \frac{12L\alpha J^2}{T}. \quad (5.5)$$

Also,

$$J^2 \leq \mathfrak{D}_p T (s^2 \ell + n_p^2 \rho). \quad (5.6)$$

Proof. Step 1: a pathwise descent inequality. For arbitrary vectors ξ_t , expanding the descent lemma gives

$$\begin{aligned} F(x_{t+1}) &\leq F(x_t) - \alpha \|g_t\|^2 - \alpha \langle g_t, \xi_t \rangle + \frac{L\alpha^2}{2} \|g_t + \xi_t\|^2 \\ &= F(x_t) - \alpha(1 - L\alpha/2) \|g_t\|^2 - \alpha(1 - L\alpha) \langle g_t, \xi_t \rangle + \frac{L\alpha^2}{2} \|\xi_t\|^2. \end{aligned}$$

As $\alpha \leq L^{-1}$, for each $k \leq T$,

$$F(x_{k+1}) - F_* + \frac{\alpha}{2} \sum_{t=1}^k \|g_t\|^2 \leq \Delta + \alpha M_k + \frac{L\alpha^2}{2} \sum_{t=1}^k \|\xi_t\|^2, \quad M_k = -(1 - L\alpha) \sum_{t=1}^k \langle g_t, \xi_t \rangle. \quad (5.7)$$

Step 2: three probability events. Apply Lemma 4.3 to the raw energy with $u = \delta/3$, and Theorem 4.1 to the noise partial sums with the same u . The two inequalities in (5.4) fail with total probability at most $2\delta/3$.

For $H > 0$, set

$$I_t = \mathbf{1}_{\{\max_{1 \leq s \leq t} (F(x_s) - F_*) \leq H\}}, \quad M_k^\circ = -(1 - L\alpha) \sum_{t=1}^k I_t \langle g_t, \xi_t \rangle.$$

These indicators are predictable. Lemma 2.7 implies $\|(1 - L\alpha)I_t g_t\| \leq \sqrt{2LH}$. The scalar increments of M° therefore have conditional moment totals bounded by $2LH Ts^2$ and $(2LH)^{p/2} T n_p^p$. Theorem 4.1, applied to the predictable stopped transform with $u = \delta/3$, gives

$$\max_{k \leq T} M_k^\circ \leq \sqrt{2LH} J \quad (5.8)$$

outside an event of probability at most $\delta/3$. Intersect the three events. No concentration inequality has been conditioned on an event involving future observations.

Step 3: removing the objective stop. Initially $F(x_1) - F_* = \Delta \leq H$. If the first exit were at x_{s+1} , $s + 1 \leq T + 1$, then $M_s = M_s^\circ$, including the increment producing that exit. Dropping the nonnegative gradient term in (5.7),

$$\begin{aligned} F(x_{s+1}) - F_* &\leq \Delta + \alpha \sqrt{2LH} J + \frac{L\alpha^2 J^2}{2} \\ &\leq \frac{H}{2} + \Delta + \frac{3}{2} L\alpha^2 J^2 = \frac{3H}{4}. \end{aligned}$$

The second inequality uses $\alpha\sqrt{2LH}J \leq H/2 + L\alpha^2J^2$. This contradicts the exit. Hence the objective remains below H throughout, and (5.8) applies to the unstopped transform on the good event.

Step 4: stationarity. At $k = T$, retaining the gradient sum in (5.7) gives

$$A_T \leq \frac{2\Delta}{\alpha T} + \frac{2\sqrt{2LH}J}{T} + \frac{L\alpha J^2}{T}.$$

Set $d_0 = \Delta/(\alpha T)$ and $e_0 = L\alpha J^2/T$. Since $H = 4\Delta + 6L\alpha^2J^2$, the middle term equals $2\sqrt{8d_0e_0 + 12e_0^2}$, which is bounded by

$$4\sqrt{2}\sqrt{d_0e_0} + 4\sqrt{3}e_0 \leq 2d_0 + (4 + 4\sqrt{3})e_0.$$

As $5 + 4\sqrt{3} < 12$, this proves (5.5). Finally, $(a + b)^2 \leq 2a^2 + 2b^2$ in (5.3) gives

$$J^2 \leq 4Ts^2\ell + 2\mathfrak{c}_p^26^{2/p}n_p^2T^{2/p}\delta^{-2/p},$$

which implies (5.6).

If $H = 0$, then $\Delta = 0$ and $J = 0$. The conditional second-moment envelope vanishes, so all errors vanish almost surely, and Lemma 2.7 makes the initial gradient zero. The recursion is constant and all conclusions hold directly. $\square$

6 Nagaev bounds under distance-dependent moments

We now replace the deterministic envelopes by Assumption 2.3. The radius in the next theorem is part of its conclusion; it is not a condition imposed on the optimizer.

Theorem 6.1 (Distance-dependent Nagaev bound). *Suppose Assumptions 2.1 and 2.3 hold. For $\delta \in (0, 1)$ define ℓ, ρ by (5.2), h_T by (3.1), and*

$$W_0 = \sigma^2 \ell + \nu_p^2 \rho, \quad W_1 = b_2 \ell + b_p \rho. \quad (6.1)$$

Let $\mathfrak{D}_p$ be the explicit constant in (5.1). If

$$0 < \alpha \leq L^{-1}, \quad h_T W_1 \leq \frac{1}{96 \mathfrak{D}_p}, \quad (6.2)$$

then, with probability at least $1 - \delta$,

$$A_T \leq \frac{6\Delta}{\alpha T} + 18 \mathfrak{D}_p L \alpha \{ \sigma^2 \log(6/\delta) + \nu_p^2 T^{2/p-1} \delta^{-2/p} \}. \quad (6.3)$$

On the same event,

$$\max_{1 \leq t \leq T+1} \|x_t - x_1\|^2 \leq R^2 := 16\alpha T \Delta + 48 \mathfrak{D}_p h_T W_0, \quad (6.4)$$

$$\max_{1 \leq t \leq T+1} \{F(x_t) - F_*\} \leq 5\Delta + 9 \mathfrak{D}_p L \alpha^2 T W_0. \quad (6.5)$$

All numerical constants are independent of d ; dependence on dimension may enter through the moment parameters.

6.1 Proof of Theorem 6.1

Proof. Step 1: deterministic localization and an auxiliary recursion. First assume $R > 0$, where R is given by (6.4). On the probability space of the original recursion, define

$$I_t = \mathbf{1}_{\{\max_{1 \leq s \leq t} \|x_s - x_1\| \leq R\}}, \quad \tilde{\xi}_t = I_t \xi_t. \quad (6.6)$$

The indicators are $\mathcal{F}_{t-1}$ -measurable. As in Lemma 4.2, $\tilde{\xi}_t$ are martingale differences, and the growth bounds imply

$$\begin{aligned} \mathbb{E}_{t-1} \|\tilde{\xi}_t\|^2 &\leq \sigma_R^2 := \sigma^2 + b_2 R^2, \\ \mathbb{E}_{t-1} \|\tilde{\xi}_t\|^p &\leq \nu_R^p, \quad \nu_R^2 := \nu_p^2 + b_p R^2. \end{aligned} \quad (6.7)$$

Indeed, on $\{I_t = 1\}$ the original iterate is within the radius, and on $\{I_t = 0\}$ the left sides vanish.

Define the auxiliary process by

$$\tilde{x}_1 = x_1, \quad \tilde{x}_{t+1} = \tilde{x}_t - \alpha \{\nabla F(\tilde{x}_t) + \tilde{\xi}_t\}. \quad (6.8)$$

This is an adapted recursion driven by the stopped errors on the original filtration. It is not an algorithm to be run, and no assertion that $\tilde{\xi}_t = e(\tilde{x}_t, Z_t)$ is needed. Proposition 5.1 requires only the martingale property and the deterministic moment envelopes, both of which hold in (6.7).

Let $\tilde{A}_T = T^{-1} \sum_{t=1}^T \|\nabla F(\tilde{x}_t)\|^2$, and let J_R be (5.3) with $s = \sigma_R$ and $n_p = \nu_R$. Proposition 5.1 yields an event $\mathcal{E}_R$ with $\mathbb{P}(\mathcal{E}_R) \geq 1 - \delta$ on which all its trajectory bounds hold for the auxiliary process. Moreover,

$$J_R^2 \leq \mathfrak{D}_p T (W_0 + W_1 R^2). \quad (6.9)$$

The radius and both noise envelopes are deterministic before applying the concentration inequality.

Step 2: the auxiliary process cannot leave the radius. On $\mathcal{E}_R$, the displacement identity for (6.8), Cauchy–Schwarz, and the maximal-noise bound give

$$\begin{aligned} \max_{k \leq T} \|\tilde{x}_{k+1} - x_1\|^2 &\leq 2\alpha^2 T^2 \tilde{A}_T + 2\alpha^2 J_R^2 \\ &\leq 8\alpha T \Delta + (24L\alpha^3 T + 2\alpha^2) J_R^2 \\ &\leq 8\alpha T \Delta + 24\mathfrak{D}_p h_T (W_0 + W_1 R^2). \end{aligned} \quad (6.10)$$

The last step uses (6.9) and $24L\alpha^3 T^2 + 2\alpha^2 T \leq 24h_T$. By the definition of R^2 , the sum of the first term and $24\mathfrak{D}_p h_T W_0$ is $R^2/2$. The remaining term is at most $R^2/4$ by (6.2). Therefore

$$\max_{k \leq T} \|\tilde{x}_{k+1} - x_1\|^2 \leq \frac{3}{4} R^2 \quad \text{on } \mathcal{E}_R. \quad (6.11)$$

Step 3: equality with the original recursion. Suppose that on $\mathcal{E}_R$ the original process first exits the closed radius- R ball at time $\tau \leq T + 1$. For every $t < \tau$, $I_t = 1$, so $\tilde{\xi}_t = \xi_t$. An induction using the identical initializations in (2.1) and (6.8) gives $\tilde{x}_t = x_t$ for every $t \leq \tau$. The increment at $t = \tau - 1$ is included in this equality. But then $\|\tilde{x}_\tau - x_1\| = \|x_\tau - x_1\| > R$, contradicting (6.11). Hence no original exit occurs, the two recursions agree up to time $T + 1$, and (6.4) follows. This is a pathwise argument on $\mathcal{E}_R$, not conditioning a martingale inequality on a future no-exit event.

Step 4: removal of the radius from the stationarity bound. On $\mathcal{E}_R$, Proposition 5.1 and (6.9) imply

$$\begin{aligned} A_T &\leq \frac{4\Delta}{\alpha T} + 12\mathfrak{D}_p L\alpha (W_0 + W_1 R^2) \\ &= \frac{4\Delta}{\alpha T} + 12\mathfrak{D}_p L\alpha W_0 + 192\mathfrak{D}_p L\alpha^2 T W_1 \Delta + 576\mathfrak{D}_p^2 L\alpha h_T W_1 W_0. \end{aligned} \quad (6.12)$$

Since $L\alpha^3 T^2 \leq h_T$,

$$192\mathfrak{D}_p L\alpha^2 T W_1 \Delta \leq 192\mathfrak{D}_p h_T W_1 \frac{\Delta}{\alpha T} \leq \frac{2\Delta}{\alpha T}.$$

Similarly,

$$576\mathfrak{D}_p^2 L\alpha h_T W_1 W_0 \leq 6\mathfrak{D}_p L\alpha W_0.$$

Substitution in (6.12) proves (6.3).

For the objective maximum, Proposition 5.1 gives

$$\begin{aligned} \max_t \{F(x_t) - F_*\} &\leq 4\Delta + 6\mathfrak{D}_p L\alpha^2 T (W_0 + W_1 R^2) \\ &= 4\Delta + 6\mathfrak{D}_p L\alpha^2 T W_0 + 96\mathfrak{D}_p L\alpha^3 T^2 W_1 \Delta + 288\mathfrak{D}_p^2 L\alpha^2 T h_T W_1 W_0 \\ &\leq 5\Delta + 9\mathfrak{D}_p L\alpha^2 T W_0, \end{aligned}$$

which is (6.5).

If $R = 0$, then $\Delta = 0$ and $W_0 = 0$. Since $\ell, \rho > 0$, the moment intercepts vanish. Lemma 2.7 gives $g_1 = 0$, and the moment bounds at x_1 give $\xi_1 = 0$ almost surely. Inductively the original process remains at x_1 and all errors vanish. The conclusions follow directly. $\square$

6.2 Trajectory control and admissibility

Theorem 6.1 allows the conditional moments to be unbounded as functions of x . Its high-probability displacement bound controls a maximum *along the run*; this is stronger than the maximum of marginal expectations in Theorem 3.1. Neither result says that the iterates remain in a fixed compact set uniformly over all horizons. For example, the radius (6.4) may grow with T because the method is permitted to travel a long distance along directions of small deterministic gradient.

The condition (6.2) is also part of the theorem. Reducing the failure probability increases both ℓ and ρ , and can force a smaller stepsize. Thus (6.3) is not obtained from a deterministic-envelope theorem by weakening an assumption while keeping the entire admissible parameter region unchanged.

When $b_2 = b_p = 0$, the feedback condition vanishes. Proposition 5.1 then gives the usual deterministic-envelope rate with a step of order $T^{-1/2}$. The more conservative distance-localized bound remains valid, but its radius is unnecessary in this special case.

7 Stepsizes, confidence levels, and output criteria

We give quantitative consequences rather than suppressing the feedback restriction in asymptotic notation. Unless otherwise indicated, $p, L, \Delta, \sigma, \nu_p, b_2, b_p$ are fixed.

7.1 A direct finite-horizon stepsize recipe

Put $c_* = (96\mathfrak{D}_p)^{-1}$. If $W_1 > 0$, the three inequalities

$$\alpha \leq \min \left\{ \frac{1}{L}, \left(\frac{c_*}{2TW_1} \right)^{1/2}, \left(\frac{c_*}{2LT^2W_1} \right)^{1/3} \right\} \quad (7.1)$$

imply (6.2), because

$$h_TW_1 = \alpha^2 TW_1 + L\alpha^3 T^2 W_1 \leq c_*/2 + c_*/2.$$

For $W_1 = 0$, only the smoothness cap remains. The parameters in (7.1) are theoretical upper bounds on noise scales and growth. The constants are explicit but conservative. For example,

$$\mathfrak{c}_4 = \frac{319}{120}, \quad \mathfrak{D}_4 = 4 + 2 \left(\frac{319}{120} \right)^2 \sqrt{6} \simeq 38.6198, \quad c_* \simeq 2.69724 \times 10^{-4}. \quad (7.2)$$

For comparison, $\mathfrak{c}_3 = 41/30$ and $\mathfrak{D}_3 \simeq 16.3345$. Thus the displayed sufficient stepsizes can be small. They certify the worst-case guarantee, not the practical stability boundary. Raw-energy and objective localization, the square-of-a-sum estimate, and the subsequent feedback absorptions all contribute slack. The unabsorbed second-moment bound in Remark 3.2 can be used when a sharper numerical certificate is needed; adaptive estimation of growth parameters is outside the present analysis.

Corollary 7.1 (An interpolating finite-horizon bound). *Suppose $\Delta > 0$ and $W_0 > 0$. Choose α equal to the minimum in (7.1) and the additional value*

$$\left(\frac{\Delta}{LTW_0} \right)^{1/2},$$

omitting growth-dependent restrictions when $W_1 = 0$. Then, with probability at least $1 - \delta$,

$$A_T \leq C_p \left\{ \frac{L\Delta}{T} + \sqrt{\frac{L\Delta W_0}{T}} + \Delta \sqrt{\frac{W_1}{T}} + \Delta \left(\frac{LW_1}{T} \right)^{1/3} \right\}, \quad (7.3)$$

where C_p depends only on p .

Proof. The choice satisfies the main theorem and $L\alpha W_0 \leq \sqrt{L\Delta W_0/T}$. Since the reciprocal of a minimum of positive numbers is the maximum of their reciprocals,

$$\frac{\Delta}{\alpha T} = \max \left\{ \frac{L\Delta}{T}, \Delta \sqrt{\frac{2W_1}{c_* T}}, \Delta \left(\frac{2LW_1}{c_* T} \right)^{1/3}, \sqrt{\frac{L\Delta W_0}{T}} \right\}.$$

Bound the maximum by the sum and apply (6.3). Since c_* and $\mathfrak{D}_p$ depend only on p , their numerical factors can be included in C_p . $\square$

When $\Delta = 0$ or $W_0 = 0$, use Theorem 6.1 and (7.1) directly, rather than dividing by a vanishing quantity. In particular, $\Delta = 0$ does not mean that the trajectory is constant when the noise intercept is nonzero.

7.2 Fixed confidence

Corollary 7.2 (Finite- p fixed-confidence rate). *Fix $\delta \in (0, 1)$ and let $\ell = \log(6/\delta)$. Define*

$$\overline{W}_1 = b_2 \ell + b_p \delta^{-2/p}.$$

If $a_0 > 0$ satisfies $a_0 \leq L^{-1}$ and

$$(a_0^2 + La_0^3)\overline{W}_1 \leq \frac{1}{96\mathfrak{D}_p}, \quad (7.4)$$

then $\alpha = a_0 T^{-2/3}$ is admissible for every positive integer T , and with probability at least $1 - \delta$,

$$A_T \leq \frac{6\Delta}{a_0} T^{-1/3} + 18\mathfrak{D}_p La_0 \sigma^2 \ell T^{-2/3} + 18\mathfrak{D}_p La_0 \nu_p^2 \delta^{-2/p} T^{2/p-5/3}. \quad (7.5)$$

Consequently $A_T = O(T^{-1/3})$ at the stated fixed confidence.

Proof. Since $p > 2$ and $T \geq 1$, $T^{2/p-1} \leq 1$, so $W_1 \leq \overline{W}_1$. Also $h_T = a_0^2 T^{-1/3} + La_0^3$. Thus (7.4) implies (6.2). Substitute the stepsize into (6.3). Finally $2/p - 5/3 < -2/3$, so the last two powers decay faster than $T^{-1/3}$. $\square$

The finite- p assumption is not needed for the exponent $1/3$ at fixed confidence: Corollary 3.3 already supplies it under second moments. The reason to impose $p > 2$ is the separation of confidence costs in (7.5) and the following increasing-confidence guarantees.

7.3 Polynomially increasing confidence

Corollary 7.3 (Increasing confidence). *Let $r > 0$, set $\delta_T = T^{-r}$ for $T \geq 2$, and define*

$$z = \frac{2(1+r)}{p} - 1, \quad w_T = \log(6T^r) + T^z. \quad (7.6)$$

If $0 < r < p - 1$, there is a sufficiently small constant $a_0 > 0$ such that

$$\alpha_T = a_0 T^{-2/3} w_T^{-1/3} \quad (7.7)$$

satisfies (6.2) for all sufficiently large T . For these runs,

$$\mathbb{P} \left\{ A_T > C \left(\frac{w_T}{T} \right)^{1/3} \right\} \leq T^{-r} \quad (7.8)$$

for a finite constant C depending only on fixed problem parameters, p, r , and a_0 . In particular,

$$0 < r \leq p/2 - 1 \implies A_T = O(T^{-1/3}(\log T)^{1/3}) \quad \text{with failure probability } T^{-r}. \quad (7.9)$$

If $p/2 - 1 < r < p - 1$, a sufficient rate is

$$A_T = O\left(T^{-[2-2(1+r)/p]/3}\right) \quad \text{with failure probability } T^{-r}. \quad (7.10)$$

Proof. Under $\delta_T = T^{-r}$, the quantity ρ in (5.2) is T^z . Write $B = b_2 + b_p$ and $S = \sigma^2 + \nu_p^2$. Then

$$W_1 \leq B w_T, \quad W_0 \leq S w_T.$$

For the choice (7.7),

$$h_T W_1 \leq B \left\{ a_0^2 \left(\frac{w_T}{T} \right)^{1/3} + L a_0^3 \right\}. \quad (7.11)$$

The condition $r < p - 1$ is equivalent to $z < 1$, and hence $w_T/T \rightarrow 0$. Choose a_0 so that $B L a_0^3 < (192\mathfrak{D}_p)^{-1}$ when $B > 0$; then for all sufficiently large T , (7.11) is at most $(96\mathfrak{D}_p)^{-1}$. If $B = 0$, the feedback condition is empty. Also $\alpha_T \rightarrow 0$, so the smoothness cap holds eventually.

The two terms in (6.3) satisfy

$$\frac{\Delta}{\alpha_T T} = \frac{\Delta}{a_0} \left(\frac{w_T}{T} \right)^{1/3}, \quad L \alpha_T W_0 \leq L a_0 S \left(\frac{w_T}{T} \right)^{2/3}.$$

Since $w_T/T \rightarrow 0$, the second is bounded by a constant times the first power of w_T/T in (7.8), even if $\Delta = 0$. This proves (7.8). If $z \leq 0$, $w_T \asymp \log T$; if $0 < z < 1$, $w_T \asymp T^z$. Substitution gives (7.9) and (7.10). $\square$

The range $r < p - 1$ is the sufficient increasing-confidence range obtained from this stepsize construction.

Example 7.4 (A fourth-moment certificate). For $p = 4$ and $\delta_T = T^{-1}$, $\rho = 1$ and $w_T = \log(6T) + 1$. A step of order $T^{-2/3}(\log T)^{-1/3}$ yields

$$A_T = O(T^{-1/3}(\log T)^{1/3}) \quad \text{with probability at least } 1 - T^{-1}.$$

At the same stepsize, Theorem 3.1 gives an expectation certificate of order $T^{-1/3}(\log T)^{1/3}$. A direct first-moment Markov conversion at failure probability T^{-1} instead gives order $T^{2/3}(\log T)^{1/3}$, which does not establish convergence. The comparison concerns the sharpness of two confidence certificates under the additional fourth moment.

7.4 Stationarity criteria and oracle complexity

Let τ be uniform on $\{1, \dots, T\}$ and independent of the complete run. Then

$$\mathbb{E}(\|\nabla F(x_\tau)\|^2 \mid x_1, \dots, x_T) = A_T, \quad \mathbb{E} \|\nabla F(x_\tau)\|^2 = \mathbb{E} A_T. \quad (7.12)$$

Corollary 3.4 gives a parameter-explicit sufficient budget for this squared-gradient output criterion. Each update uses exactly one oracle call; the expectation guarantee already implies the first-moment stationarity criterion used in the external BG-0 lower bound. Full-gradient evaluations used to report A_T are not part of the optimizer and are unnecessary for selecting x_τ .

For a probability bound, if $\mathbb{P}(A_T > B) \leq \delta$ and $u \in (0, 1)$, conditional Markov and (7.12) give

$$\mathbb{P}\{\|\nabla F(x_\tau)\|^2 > B/u\} \leq \delta + u. \quad (7.13)$$

Indeed, on $\{A_T \leq B\}$ the conditional probability is at most u , and the complementary event has probability at most δ . Also $\min_{t \leq T} \|\nabla F(x_t)\|^2 \leq A_T$ pathwise. Neither observation is a last-iterate theorem or an assertion that the iterate minimizing the full gradient norm is available without additional computation.

8 Objective-gap moment growth and a root- T refinement

Distance growth alone allows noise to increase along flat objective directions. When the noise scale is controlled directly by the objective gap, a sharper localization condition is available. This is an additional assumption, not a consequence of the two-point Lipschitz condition by itself.

Assumption 8.1 (Objective-gap moment growth). For $p > 2$ there are deterministic $\omega_2, \omega_p, \gamma_2, \gamma_p \geq 0$ such that, almost surely,

$$\begin{aligned} \mathbb{E}_{t-1} \|\xi_t\|^2 &\leq \omega_2 + \gamma_2 \{F(x_t) - F_*\}, \\ (\mathbb{E}_{t-1} \|\xi_t\|^p)^{2/p} &\leq \omega_p + \gamma_p \{F(x_t) - F_*\}. \end{aligned} \quad (8.1)$$

The intercepts ω_2, ω_p are squared moment scales; a_0 denotes a horizon-dependent stepsize coefficient. For this assumption the finite-horizon integrability argument remains valid: the smooth upper estimate

$$F(x) - F_* \leq \Delta + \|\nabla F(x_1)\| \|x - x_1\| + \frac{L}{2} \|x - x_1\|^2$$

converts the squared root-moment bound to a constant times $1 + \|x - x_1\|^2$, so Lemma 2.6's induction applies.

Theorem 8.2 (Nagaev bound under objective-gap growth). *Suppose Assumptions 2.1 and 8.1 hold. Define*

$$Q_0 = \omega_2 \ell + \omega_p \rho, \quad Q_1 = \gamma_2 \ell + \gamma_p \rho, \quad (8.2)$$

with ℓ, ρ as in (5.2). If

$$0 < \alpha \leq L^{-1}, \quad L\alpha^2 T Q_1 \leq \frac{1}{24\mathfrak{D}_p}, \quad (8.3)$$

then, with probability at least $1 - \delta$,

$$A_T \leq \frac{8\Delta}{\alpha T} + 18\mathfrak{D}_p L\alpha Q_0, \quad (8.4)$$

$$\max_{1 \leq t \leq T+1} \{F(x_t) - F_*\} \leq H := 8\Delta + 12\mathfrak{D}_p L\alpha^2 T Q_0. \quad (8.5)$$

Proof. Assume $H > 0$ and define predictable indicators on the original trajectory by

$$I_t = \mathbf{1}_{\{\max_{1 \leq s \leq t} (F(x_s) - F_*) \leq H\}}, \quad \tilde{\xi}_t = I_t \xi_t.$$

Their deterministic envelopes are $s_H^2 = \omega_2 + \gamma_2 H$ and $n_H^2 = \omega_p + \gamma_p H$. Drive the auxiliary recursion (6.8) with these errors. Proposition 5.1 provides a good event of probability at least $1 - \delta$ and a deterministic J_H obeying

$$J_H^2 \leq \mathfrak{D}_p T (Q_0 + Q_1 H).$$

On this event,

$$\begin{aligned} \max_t \{F(\tilde{x}_t) - F_*\} &\leq 4\Delta + 6\mathfrak{D}_p L\alpha^2 T (Q_0 + Q_1 H) \\ &= \frac{H}{2} + 6\mathfrak{D}_p L\alpha^2 T Q_1 H \leq \frac{3H}{4}. \end{aligned}$$

If the original trajectory first exited the objective level H at time τ , the errors and recursions would agree through the update producing x_τ . Then $F(x_\tau) = F(\tilde{x}_\tau) \leq F_* + 3H/4$, a contradiction. The trajectories therefore agree on the good event and (8.5) follows.

Their stationarity bound satisfies

$$\begin{aligned} A_T &\leq \frac{4\Delta}{\alpha T} + 12\mathfrak{D}_p L\alpha (Q_0 + Q_1 H) \\ &= \frac{4\Delta}{\alpha T} + 12\mathfrak{D}_p L\alpha Q_0 + 96\mathfrak{D}_p L\alpha Q_1 \Delta + 144\mathfrak{D}_p^2 L^2 \alpha^3 T Q_1 Q_0. \end{aligned}$$

By (8.3), the last two terms are at most $4\Delta/(\alpha T)$ and $6\mathfrak{D}_p L\alpha Q_0$, respectively. This proves (8.4). If $H = 0$, then $\Delta = Q_0 = 0$, so the gradient and the noise at initialization vanish. Induction gives a constant trajectory, proving the degenerate case. $\square$

Corollary 8.3 (Fixed-confidence root- T rate). *Fix $\delta \in (0, 1)$ and all problem parameters in Assumption 8.1. A sufficiently small fixed $a_0 > 0$ and $\alpha = a_0 T^{-1/2}$ satisfy (8.3) for all sufficiently large T . For these runs,*

$$A_T = O(T^{-1/2}) \quad \text{with probability at least } 1 - \delta. \quad (8.6)$$

Proof. For $T \geq 1$, $Q_1 \leq \gamma_2 \ell + \gamma_p \delta^{-2/p}$. Hence

$$L\alpha^2 T Q_1 \leq L a_0^2 (\gamma_2 \ell + \gamma_p \delta^{-2/p}),$$

which can be made at most $(24\mathfrak{D}_p)^{-1}$. The smoothness cap eventually holds. Substituting in (8.4), the transient and variance terms have order $T^{-1/2}$ and the polynomial term has order $T^{2/p-3/2} = o(T^{-1/2})$. $\square$

Proposition 8.4 (A geometric sufficient condition). *Suppose the pointwise iid-oracle version of Assumption 2.3 holds. If deterministic $r_0, \kappa \geq 0$ satisfy*

$$\|x - x_1\|^2 \leq r_0^2 + \kappa \{F(x) - F_*\}, \quad x \in \mathbb{R}^d, \quad (8.7)$$

then Assumption 8.1 holds with

$$\omega_2 = \sigma^2 + b_2 r_0^2, \quad \omega_p = \nu_p^2 + b_p r_0^2, \quad \gamma_2 = b_2 \kappa, \quad \gamma_p = b_p \kappa.$$

Alternatively, if a minimizer x_ satisfies $F(x) - F_* \geq (\mu/2) \|x - x_*\|^2$ for $\mu > 0$, and $\|e(x, Z)\|_q \leq s_{q,*} + K_q \|x - x_*\|$ for $q = 2, p$, then one may take*

$$\omega_q = 2s_{q,*}^2, \quad \gamma_q = 4K_q^2/\mu, \quad q = 2, p. \quad (8.8)$$

Proof. Substitute (8.7) into the two pointwise squared root-moment bounds, and use the fresh-sample identity (2.8). For the second assertion,

$$\|e(x, Z)\|_q^2 \leq 2s_{q,*}^2 + 2K_q^2 \|x - x_*\|^2 \leq 2s_{q,*}^2 + \frac{4K_q^2}{\mu} \{F(x) - F_*\}.$$

Conditioning gives (8.8). $\square$

Strong convexity is sufficient for the second geometric inequality when F_* is the minimum, but is not necessary. Neither ordinary nonconvex smoothness nor a bound on the objective alone implies (8.7). Also, a quadratic growth inequality relative to the *distance to a set* of minimizers does not automatically bound distance to a particular point of that set.

Relation to expected smoothness. In the iid setting, Assumption 8.1 at $q = 2$ gives

$$\mathbb{E}_Z \|G(x, Z)\|^2 \leq \|\nabla F(x)\|^2 + \omega_2 + \gamma_2 \{F(x) - F_*\}.$$

This is a special case of the ABC condition of Khaled and Richtárik (2023). Theorem 8.2 adds a specified p th-moment growth assumption and an explicit variance/rare-shock confidence bound. Its contribution is the finite-moment confidence decomposition under this established second-moment framework.

9 Examples and the necessity of noise-sensitive stability

All random observations in this section are independent across iterations. The examples verify the assumptions pointwise in the parameter before conditional substitution along the trajectory.

9.1 Random affine gradient errors

Example 9.1 (Matrix-valued multiplicative noise). Let $B = B(Z)$ be a random $d \times d$ matrix and $\eta = \eta(Z)$ a random vector, with $\mathbb{E}B = 0$, $\mathbb{E}\eta = 0$, $\|B\|_{\text{op}}\|_p < \infty$, and $\|\eta\|_p < \infty$. No independence between B and η is required. Define

$$G(x, Z) = \nabla F(x) + B(Z)(x - x_1) + \eta(Z). \quad (9.1)$$

For $q = 2, p$, Minkowski gives

$$\|e(x, Z)\|_q \leq \|\eta\|_q + \|B\|_{\text{op}}\|_q \|x - x_1\|.$$

Thus the squared root-moment growth constants may be chosen as

$$\sigma^2 = 2\|\eta\|_2^2, \quad b_2 = 2\|B\|_{\text{op}}\|_2^2, \quad \nu_p^2 = 2\|\eta\|_p^2, \quad b_p = 2\|B\|_{\text{op}}\|_p^2.$$

The full stochastic gradient is L^p -Lipschitz with constant at most $L + \|B\|_{\text{op}}\|_p$. If B is symmetric, the oracle is the gradient of

$$f(x, Z) = F(x) + \frac{1}{2}(x - x_1)^\top B(Z)(x - x_1) + \eta(Z)^\top (x - x_1).$$

The objective is $\mathbb{E}f(x, Z) = F(x)$. Individual random losses are not required to be bounded below. If there is a direction v with $\|Bv\|_q > 0$, the reverse triangle inequality along $x = x_1 + rv$ gives $\|e(x, Z)\|_q \geq |r| \|Bv\|_q - \|\eta\|_q$, so a global moment envelope is impossible.

9.2 Nonconvex flat directions: distance growth without gap growth

Example 9.2 (An objective level cannot control the noise). Let Z be centered, nondegenerate, and in L^p . On $\mathbb{R}^2$, take

$$F(u, v) = 1 - \cos u, \quad f((u, v), Z) = 1 - \cos u + Zuv, \quad x_1 = (\pi/2, 0). \quad (9.2)$$

Then $F_* = 0$, $L = 1$, and F is nonconvex because its Hessian is $\text{diag}(\cos u, 0)$. The oracle and its error are

$$G((u, v), Z) = (\sin u + Zv, Zu), \quad e((u, v), Z) = Z(v, u).$$

Consequently

$$\|e((u, v), Z)\|_q = \|Z\|_q \sqrt{u^2 + v^2}, \quad q = 2, p, \quad (9.3)$$

and G is L^p -Lipschitz with constant at most $1 + \|Z\|_p$. Relative to x_1 , squared root-moment growth holds with intercept $2\|Z\|_q^2(\pi/2)^2$ and coefficient $2\|Z\|_q^2$. Theorem 6.1 therefore applies for its stated stepsizes.

At $(u, v) = (0, v)$, however, $F(0, v) - F_* = 0$ while $\|e((0, v), Z)\|_q = |v| \|Z\|_q$. No finite constants ω_q, γ_q can satisfy the global pointwise gap-growth inequality. The example has a bounded objective, no coercivity, and unbounded noise moments along a minimum-level set. It separates the two localization theorems without relying on an abstract counterexample to compactness.

9.3 A nonconvex coercive example with gap growth

Example 9.3 (Quadratic growth without convexity). Let

$$F(x) = \frac{x^2}{2} + 2(1 - \cos x), \quad x \in \mathbb{R}. \quad (9.4)$$

Here $F_* = 0$, $F''(x) = 1 + 2\cos x$, and $L = 3$ is a valid smoothness constant. Since $F''(\pi) = -1$, the objective is nonconvex. Nevertheless, $F(x) \geq x^2/2$ and

$$|x - x_1|^2 \leq 2x^2 + 2x_1^2 \leq 4F(x) + 2x_1^2.$$

Thus (8.7) holds with $r_0^2 = 2x_1^2$ and $\kappa = 4$. Any affine-noise oracle in Example 9.1 satisfies the gap-growth theorem, and sufficiently small steps of order $T^{-1/2}$ yield the fixed-confidence root- T bound. Convexity is not used in this verification.

9.4 Random-design regression with a nonconvex penalty

Example 9.4 (Unbounded random design). Let $(a, y) \in \mathbb{R}^d \times \mathbb{R}$ satisfy

$$\mathbb{E} \|a\|^{2p} < \infty, \quad \mathbb{E}(|y| \|a\|)^p < \infty, \quad \mathbb{E} y^2 < \infty, \quad p > 2. \quad (9.5)$$

For $\lambda \geq 0$, consider

$$\begin{aligned} F(x) &= \frac{1}{2} \mathbb{E} (a^\top x - y)^2 + \lambda \sum_{j=1}^d (1 - \cos x_j), \\ G(x, (a, y)) &= a(a^\top x - y) + \lambda \sin x, \end{aligned} \quad (9.6)$$

where the sine is coordinatewise. Put $Q = \mathbb{E} a a^\top$ and $c = \mathbb{E} a y$. The moment assumptions imply these quantities are finite and justify differentiating the quadratic expectation directly by expansion. Thus $\nabla F(x) = Qx - c + \lambda \sin x$, $F \geq 0$, and $L = \|Q\|_{\text{op}} + \lambda$ is a valid smoothness bound whenever positive.

Write

$$B = a a^\top - Q, \quad \eta = Bx_1 - (ay - c).$$

Then $\mathbb{E} B = 0$, $\mathbb{E} \eta = 0$, and the error equals $B(x - x_1) + \eta$. The moment conditions (9.5) imply the L^p operator-norm and vector-moment conditions in Example 9.1. Indeed, $\|a a^\top\|_{\text{op}} = \|a\|^2$, and Minkowski bounds both B and η . No almost-sure boundedness of the design or response is required.

This model can be genuinely nonconvex: in dimension one with $Q = 1$ and $\lambda = 2$, the Hessian is $1 + 2\cos x$, which is negative at $x = \pi$. The error is nevertheless unbiased and stochastic Lipschitz continuity holds. Applying the distance-growth theorem does not require a smallest eigenvalue bound on Q or coercivity of this objective.

9.5 A smoothness-only stepsize cap is insufficient

Proposition 9.5 (Multiplicative instability). *There is a one-dimensional strongly convex objective and an unbiased iid stochastic-gradient oracle that is L^p -Lipschitz for every finite p , but whose SGD iterates diverge on every sample path at a stepsize satisfying $\alpha \leq (8L)^{-1}$.*

Proof. Let $F(x) = x^2/2$, $x_1 = 1$, and let Z_t be iid symmetric signs. Define

$$G(x, Z) = (1 + 24Z)x, \quad \alpha = 1/8.$$

Then $L = 1$, $\mathbb{E}G(x, Z) = x$, $G(0, Z) = 0$, and

$$\|G(x, Z) - G(y, Z)\|_p = \|1 + 24Z\|_p |x - y| < \infty$$

for every finite p . The recursion is

$$x_{t+1} = \left(\frac{7}{8} - 3Z_t\right) x_t.$$

The two multipliers are $-17/8$ and $31/8$. Therefore

$$|x_t| \geq (17/8)^{t-1}, \quad A_T = \frac{1}{T} \sum_{t=1}^T x_t^2 \geq \frac{1}{T} (17/8)^{2(T-1)}$$

on every sample path. This establishes the assertion. $\square$

More generally, with $G(x, Z) = (1 + cZ)x$ and symmetric signs,

$$\mathbb{E}x_{t+1}^2 = \{(1 - \alpha)^2 + c^2\alpha^2\} \mathbb{E}x_t^2. \quad (9.7)$$

Mean-square contraction holds exactly when $0 < \alpha < 2/(1 + c^2)$, by expanding the factor and requiring it to be less than one. This calculation establishes the need for noise-sensitive stability, rather than an asymptotic rate lower bound for this example. The broad BG-0 minimax conclusion instead follows from the oracle-class comparison in Section 3.1.

10 An additive rare-shock lower bound

The BG-0 oracle-complexity lower bound in Section 3.1 and the confidence lower bound below concern different questions. The next elementary construction shows that unchanged SGD cannot have a variance-only logarithmic confidence certificate over a finite- p class. It specializes the sparse-shock argument in the supplied manuscript identified in Section 11.1; the full calculation is included, and no separate originality claim is made for it. The same broad rare-event mechanism underlies the Nagaev sharpness phenomenon for linear averaged SGD in Zhu et al. (2022).

Proposition 10.1 (Polynomial confidence cost for unchanged SGD). *Let $F(x) = \lambda x^2/2$ with $\lambda > 0$, $x_1 = 0$, $p > 2$, and $\nu > 0$. Suppose*

$$\vartheta = \alpha\lambda \in (0, 1/4], \quad N = \left\lfloor \frac{1}{2\vartheta} \right\rfloor,$$

and let T be even with $T \geq 2N$. For every $\delta \in (0, 1/8]$, there is a centered iid additive-noise law with $\mathbb{E}|\xi_t|^p = \nu^p$ such that

$$\mathbb{P} \left\{ A_T \geq \frac{\lambda\alpha\nu^2}{16 \cdot 4^{2/p}} T^{2/p-1} \delta^{-2/p} \right\} \geq \delta. \quad (10.1)$$

The law may depend on T and δ . Its variance is

$$\mathbb{E}\xi_t^2 = \nu^2 (4\delta/T)^{1-2/p}. \quad (10.2)$$

Proof. Set $r_0 = 4\delta/T$ and $v = \nu r_0^{-1/p}$. Independently at each time, let

$$\mathbb{P}(\xi_t = v) = \mathbb{P}(\xi_t = -v) = r_0/2, \quad \mathbb{P}(\xi_t = 0) = 1 - r_0.$$

The law is centered, $\mathbb{E}|\xi_t|^p = r_0 v^p = \nu^p$, and (10.2) follows from $r_0 v^2$.

Let $\mathcal{E}$ be the event that there is exactly one nonzero shock among times $1, \dots, T$, located in $1, \dots, T/2$. Then

$$\begin{aligned} \mathbb{P}(\mathcal{E}) &= \frac{T}{2} r_0 (1 - r_0)^{T-1} = 2\delta (1 - 4\delta/T)^{T-1} \\ &\geq 2\delta \{1 - (T-1)4\delta/T\} \geq 2\delta(1 - 4\delta) \geq \delta. \end{aligned} \quad (10.3)$$

We used Bernoulli's inequality and $\delta \leq 1/8$.

If the unique shock occurs at time $s \leq T/2$, the quadratic recursion gives

$$x_{s+1+j} = -\alpha \xi_s (1 - \vartheta)^j, \quad j \geq 0 \quad (10.4)$$

as long as there are no later shocks, while $x_1 = \dots = x_s = 0$. Since $\vartheta \leq 1/4$, $1/(2\vartheta) \geq 2$ and

$$N = \lfloor 1/(2\vartheta) \rfloor \geq 1/(4\vartheta).$$

For $0 \leq j < N$, Bernoulli's inequality gives $(1 - \vartheta)^j \geq 1 - \vartheta j \geq 1/2$. All N iterates in (10.4) with these indices enter A_T , because $s + N \leq T$. Therefore, on $\mathcal{E}$,

$$A_T \geq \frac{N \lambda^2 \alpha^2 v^2}{4T} \geq \frac{\lambda \alpha v^2}{16T} = \frac{\lambda \alpha \nu^2}{16 \cdot 4^{2/p}} T^{2/p-1} \delta^{-2/p}.$$

Combine this inequality with (10.3). □

Remark 10.2 (Exact finite-horizon response). On a path with its only shock of magnitude v at time s , summing the geometric series in (10.4) gives

$$A_T = \frac{\lambda \alpha v^2}{T(2 - \vartheta)} \left\{ 1 - (1 - \vartheta)^{2(T-s)} \right\}. \quad (10.5)$$

Indeed, before simplification the expression is $\lambda^2 \alpha^2 v^2 T^{-1} \sum_{j=0}^{T-s-1} (1 - \vartheta)^{2j}$, whose denominator is $1 - (1 - \vartheta)^2 = \vartheta(2 - \vartheta)$. When $s = T$ the sum is empty and both sides vanish. This indexing matters because A_T uses the gradients before the updates.

The additive oracle $G(x, Z) = \lambda x + Z$ has stochastic Lipschitz constant λ independent of the noise law, and the actual centered error has $b_2 = b_p = 0$. Thus Proposition 10.1 applies to a subclass already covered by the main theorem. For the law above, the ratio of the variance contribution to the polynomial contribution in a Nagaev bound is

$$\frac{\nu^2 (4\delta/T)^{1-2/p} \log(6/\delta)}{\nu^2 T^{2/p-1} \delta^{-2/p}} = 4^{1-2/p} \delta \log(6/\delta),$$

which tends to zero as $\delta \downarrow 0$. Consequently a logarithmic variance term alone cannot cover the polynomial confidence scale uniformly over this class. The construction changes its law with the horizon and confidence; it is not a fixed-law tail asymptotic. It also gives no lower bound forcing the distance-feedback stepsize in (6.2).

11 Discussion and scope

The results separate stability, oracle complexity, and confidence. A second-moment descent–displacement closure is enough for ordinary single-sample SGD to attain the minimax stochastic complexity of the smooth BG-0 class. Higher moments serve a different purpose: they separate the logarithmic variance cost from the polynomial cost of rare shocks. The localization radius in Theorem 6.1 follows from the same recursion that determines stationarity, so the proof does not require bounded iterates or modify the update.

The optimality comparison is made in the second-moment BG-0 oracle model. Two-point stochastic Lipschitz continuity is a sufficient but stronger condition, and its common-sample regularity can be used by variance-reduced methods. Likewise, objective-gap growth is stronger in a different direction: it permits localization with $L\alpha^2T$ instead of h_T and hence a root- T fixed-confidence guarantee. The flat-direction and coercive examples demonstrate these distinctions.

Scope. The guarantees concern average squared gradients and the explicitly stated randomized-output conversions, not last iterates or convergence to a specified minimizer. They hold for a fixed horizon with a deterministic stepsize that is constant during that run; they do not establish a single horizon-free infinite-run result. Confidence-sensitive stepsizes use upper bounds on the growth parameters. Dimension-free numerical constants still allow dimensional dependence through those parameters. The finite- p confidence exponents and localization restrictions are sufficient bounds, whereas the BG-0 minimax comparison concerns the second-moment stochastic oracle complexity. The sparse-shock law in Section 10 may vary with horizon and confidence. This theoretical paper reports no simulation evidence or empirical estimate of a worst-case rate.

Directions suggested by the analysis. A horizon-free version of the descent–displacement closure, sharper constants for finite-sample use, and a direct plain-SGD analysis exploiting two-point stochastic smoothness are natural next questions. They are distinct from the fixed-horizon BG-0 complexity established here. Data-driven control of the state-dependent moment coefficients would also make the confidence-sensitive stepsize rule more useful in applications.

11.1 Acknowledgment of supplied related work

The deterministic-envelope argument in Section 5 and the quadratic sparse-shock construction in Section 10 were motivated by the supplied manuscript *Nagaev Bounds for Nonconvex Momentum: Finite Moments and Temporal Dependence*, specifically its Theorems 2.1–2.2 and Appendices A–B. These ingredients are reproduced with full proofs and are not counted among the new contributions. The supplied version is anonymized; no author identity or publication status is assigned here. The mathematical arguments above use public concentration results and are self-contained apart from the expressly cited probability and oracle lower-bound theorems.

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